Emil Mammadli

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Experience

AZERCOSMOS | Flight Dynamics & Attitude Control Intern
Aug 2022 – Sep 2022, Sep 2024 - Oct 2024 | Gobustan, Azerbaijan

- Gained expertise in satellite communication fundamentals. Created SatTool for orbit propagation, eclipse prediction, and collision probability estimation. Developed Telemetry, a GPS tracking system with mobile/web clients and real-time data visualization. Analyzed utilization of quantum computing algorithms in orbital perturbation analysis.

TÜBITAK | CFD Researcher
November 2024 – May 2025 | Ankara, Türkiye

- Applied CFD knowledge in two research research projects. Utilized SolidWorks for domain geometry design, ANSYS for meshing process, and OpenFOAM as the numerical solver.

Research

IMPLEMENTATION OF LOW-DISSIPATION CENTRAL-UPWIND SCHEMES IN OPENFOAM | OpenFOAM

- Development of a novel OpenFOAM solver using new low-dissipation central-upwind schemes.

MODELING THE ELECTROMAGNETIC WAVE PROPAGATION IN OPENFOAM | OpenFOAM

- Development of a new OpenFOAM solver using Maxwell's equation of wave propagation.

AXISYMMETRIC FLOW ANALYSIS AND GEOMETRY OPTIMIZATION OF LAUNCH PADS VIA DISCRETE ADJOINT METHOD | OpenFOAM, Ansys Meshing, Python

- An effective approach to optimization problem for the launch pad geometry is proposed. The simulation is done in 2D axisymmetric configuration for computational efficiency. DAFoam is utilized for the design optimization.

SHAPE OPTIMIZATION OF UNSTEADY SUPERSONIC FLOWS USING A DISCRETE ADJOINT METHOD IN OPENFOAM | OpenFOAM

- Development of unsteady density-based solver in OpenFOAM via DAFoam framework, for discrete-adjoint based method optimization.

Personal Projects

FINITE ELEMENT METHOD SOLVER WITH PARALLEL PROCESSING, MODAL ANALYSIS & OPTIMIZATION WITH DISCRETE ADJOINT METHOD | Python, C++

- FEM code with parallel computing and shape optimization routine via discrete adjoint method to get the desired natural frequencies through modal analysis.

OPENFFD | Python, C++

- Advanced free-form deformation generator for computational design; finite-difference method optimization with several test cases (airfoil, cylinder).

CEA ANALYZER | Solidworks, OpenFOAM, Ansys Meshing, OpenRocket

- Analyzing and filtering CEA output data, choosing best fuel properties, and using the method of characteristics to develop nozzle geometry.

Skills

LANGUAGES

Proficient:
English • Russian • Azerbaijani
Turkish

Elementary:
French • Ukrainian

TECHNICAL

Programming:
C++ • JavaScript • Matlab
Fortran • Python

Engineering:
ANSYS • OpenFOAM •
Siemens NX • SolidWorks

Certificates

IELTS
Overall score: 8.5

SAT
Overall score: 1490

Education

CONCORDIA UNIVERSITY
Master's in Aerospace Engineering
Jul 2026 - Present | Montreal, CA

MIDDLE EAST TECHNICAL UNIVERSITY
Bachelor's in Aerospace Engineering
Grad Jun 2026 | Ankara, TR